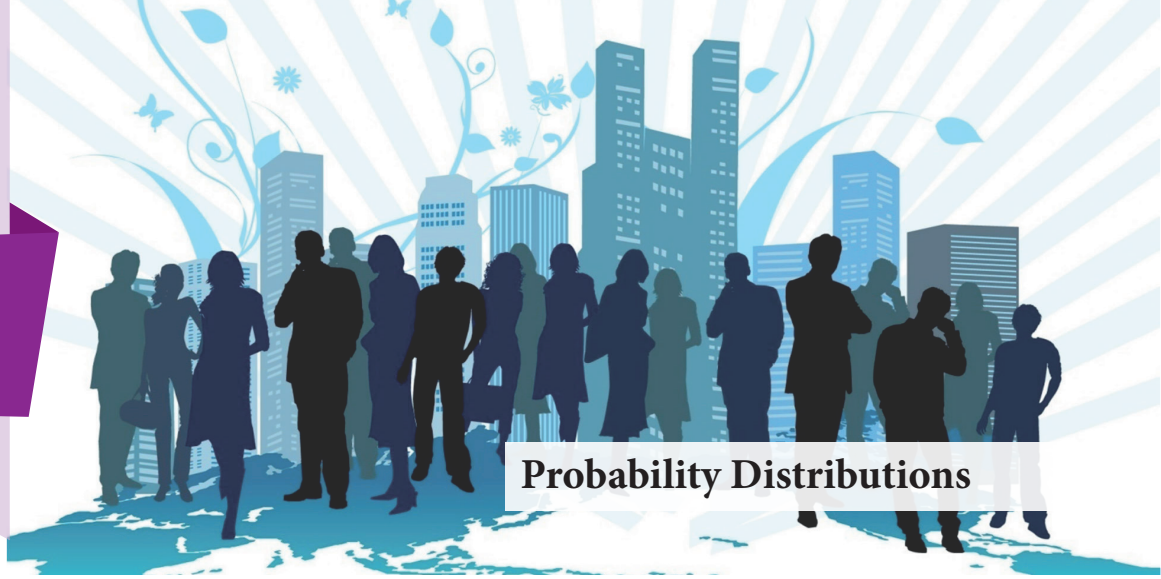
Chapter

10



Probability Distributions



Jacob Bernoulli
(6 Jan'1655 – 16 Aug'1705)

Jacob Bernoulli was born in Basel, Switzerland, is a mathematician. He contradicted to the desires of his parents, studied mathematics and astronomy and made tremendous achievement in later years. He traveled throughout Europe from 1676 to 1682, learning about the latest discoveries in mathematics and the sciences.

Jacob Bernoulli's first important contributions were a pamphlet on the parallels of logic and algebra published in 1685, after he worked on probability and geometry in 1687 . Jacob Bernoulli's paper of 1690 is important for the history of calculus, and in that integration was used with its complete meaning. In 1683 Bernoulli discovered the constant e by studying a question about compound interest which required him to find the value of the following expression $(1 + 1/n)^{1/n}$ (which is in fact e).

'Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write'. ***– H.G. Wells***

Learning Objectives

BD8P7A

- Understands probability distributions
- Differentiates discrete and continuous distribution
- Estimates value using binomial, poisson distribution
- Calculates the values using uniform, normal distribution
- Applies the method of fitting various distributions.